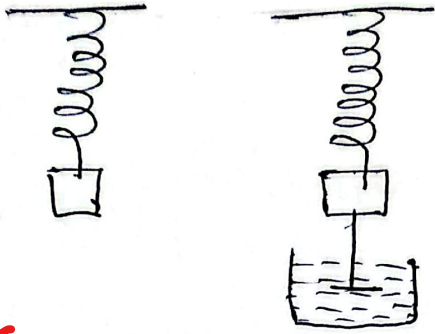


PHY-L-09-17 oct

Damped Oscillation



$$F_x - v$$

$$F' = -bv$$

Sp

$$F = ma = -kx - bv$$

$$\Rightarrow \frac{d^2x}{dt^2} \times m = -kx - b \frac{dx}{dt}$$

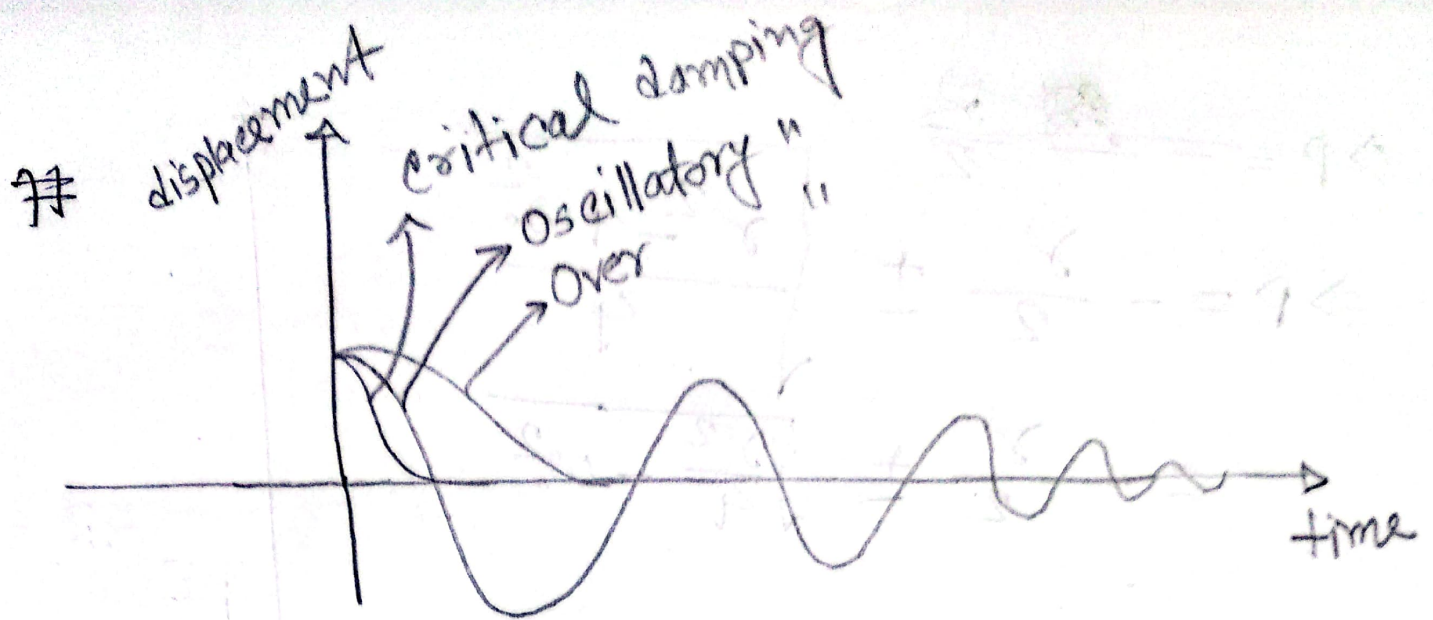
$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}x - \frac{b}{m} \frac{dx}{dt}$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\omega^2 x - \gamma \frac{dx}{dt}$$

$$\Rightarrow \frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega^2 x = 0$$

$$\left[\begin{array}{l} \omega^2 = \frac{k}{m} \\ \gamma = \frac{b}{m} \end{array} \right. \rightarrow \text{damping constant}$$





$\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$

Let, $x = B e^{pt}$ [~~For oscillatory damping~~]

$\Rightarrow \frac{dx}{dt} = B p e^{pt}$

$\Rightarrow \frac{d^2x}{dt^2} = B p^2 e^{pt}$

Now, $B p^2 e^{pt} + \gamma B p e^{pt} + \omega_0^2 B e^{pt} = 0$

$\Rightarrow p^2 + \gamma p + \omega_0^2 = 0$ [divide by $B e^{pt}$]

$\therefore p = \frac{-\gamma \pm \sqrt{\gamma^2 - 4 \cdot 1 \cdot \omega_0^2}}{2}$

~~$$\Rightarrow p = -\frac{\gamma}{2} \pm \frac{\gamma^2}{2}$$~~

$$\Rightarrow p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2 - 4\omega_0^2}{4}}$$

$$= -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

if, $\omega_0^2 > \frac{\gamma^2}{4} \rightarrow$ Oscillatory damping

$\omega_0^2 = \frac{\gamma^2}{4} \rightarrow$ critical

$\omega_0^2 < \frac{\gamma^2}{4} \rightarrow$ Over

Case 1: $(\omega_0^2 > \frac{\gamma^2}{4})$

$$\sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

$$= \sqrt{-\left(\omega_0^2 - \frac{\gamma^2}{4}\right)}$$

$$= \sqrt{-\omega_1^2} \quad \left[\omega_1^2 = \omega_0^2 - \frac{\gamma^2}{4} \right]$$

$$= \sqrt{\omega_1^2 j^2}$$

$$= \pm \omega_1 j$$

$$\therefore p = -\frac{\gamma}{2} \pm \omega_1 j$$

$$\therefore p_1 = -\frac{\gamma}{2} + \omega_1 j$$

$$p_2 = -\frac{\gamma}{2} - \omega_1 j$$

Now, $x = B e^{pt}$

$$= B_1 e^{p_1 t} + B_2 e^{p_2 t}$$

$$= B_1 e^{(-\frac{\gamma}{2} + j\omega_1)t} + B_2 e^{(-\frac{\gamma}{2} - j\omega_1)t}$$

$$= B_1 e^{-\frac{\gamma t}{2}} e^{j\omega_1 t} + B_2 e^{-\frac{\gamma t}{2}} e^{-j\omega_1 t}$$

$$= e^{-\frac{\gamma t}{2}} (B_1 e^{j\omega_1 t} + B_2 e^{-j\omega_1 t})$$

$$\therefore x = A e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \phi)$$

→ to plot in the graph:

$$y = 5 e^{-3x} \cos(7x)$$

$$\left| \begin{array}{l} x = y \\ t = x \\ A = 5 \\ \frac{\gamma}{2} = 3 \\ \omega_1 = 7 \end{array} \right.$$

case 2: $\left(\frac{\gamma^2}{4} = \omega_0^2 \right)$

$$p = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

$$= -\frac{\gamma}{2} \left[\omega_0^2 = \frac{\gamma^2}{4} \right]$$

$$\therefore x = B e^{-\frac{\gamma}{2} t}$$

→ to plot in the graph:

$$y = 2 e^{-5x}$$

$$\left| \begin{array}{l} x = y \\ t = x \\ B = 2 \\ \gamma/2 = 5 \end{array} \right.$$

PHY - L - 10 - 21 Oct

case (iii) $(\omega_0^2 < \frac{\gamma^2}{4})$

(overdamp)

$$\therefore P = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

$$= -\frac{\gamma}{2} + \lambda$$

$$\therefore P_1 = -\frac{\gamma}{2} + \lambda, P_2 = -\frac{\gamma}{2} - \lambda$$

$$\therefore x = B_1 e^{P_1 t} + B_2 e^{P_2 t}$$

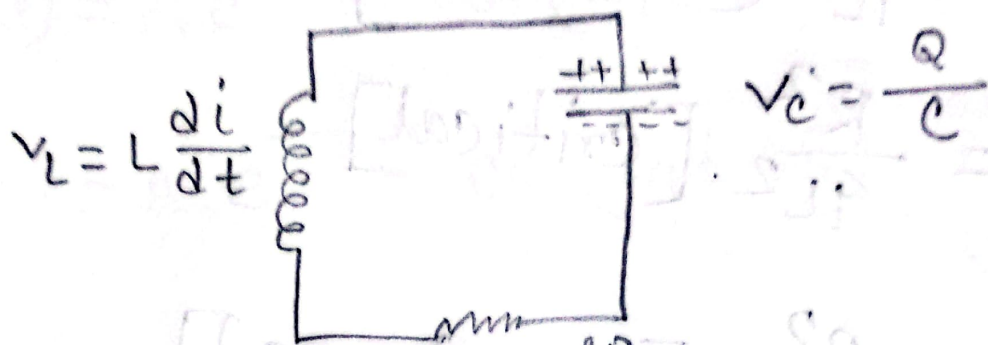
$$= B_1 e^{(-\frac{\gamma}{2} + \lambda)t} + B_2 e^{(-\frac{\gamma}{2} - \lambda)t}$$

critical: $x(t) = (A + Bt) e^{-\frac{\gamma}{2}t}$

oscillatory:

$$x(t) = A e^{-\frac{\gamma}{2}t} \cos(\omega_1 t + \phi)$$

RLC Circuit



Resistor (absorb energy each cycle)

$$\Rightarrow iR + L \frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{1}{LC} Q = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \gamma \frac{dQ}{dt} + \omega_0^2 Q = 0$$

$$\left| \begin{array}{l} \gamma = \frac{R}{L} \\ \omega_0^2 = \frac{1}{LC} \end{array} \right.$$

RLC condition for damped

$$\omega_0^2 > \frac{\gamma^2}{4} \text{ [Osc]}$$

$$\frac{1}{LC} > \frac{R^2}{4L^2} \text{ [Osc]}$$

RLC circuit

$$\frac{1}{LC} > \frac{R^2}{4L^2} [\text{oscillatory}]$$

$$\frac{1}{LC} = \frac{R^2}{4L^2} [\text{critical}]$$

$$\frac{1}{LC} < \frac{R^2}{4L^2} [\text{overdamped}]$$

Mass spring system



$$0 = m \ddot{x} + b \dot{x} + kx$$
$$0 = m s^2 x + b s x + k x$$
$$0 = (ms^2 + bs + k)x$$
$$ms^2 + bs + k = 0$$
$$s = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

Condition for overdamped

$$\left[\frac{b^2}{4m^2} > \frac{k}{m} \right]$$
$$\left[\frac{b^2}{4m^2} < \frac{k}{m} \right]$$
$$\left[\frac{b^2}{4m^2} = \frac{k}{m} \right]$$

case i: $\left(\frac{1}{LC} > \frac{R^2}{4L^2}\right)$

$$Q(t) = A e^{-\frac{Rt}{2L}} \cos(\omega_1 t + \phi)$$

$$= A e^{-\frac{Rt}{2L}} \cos(\omega_1 t + \phi)$$

Resonant angular frequency
 $\omega_1^2 = \omega_0^2 - \frac{R^2}{4L^2}$

$\omega_1 = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$
 angular frequency of oscillation

Exm

$C = 1 \mu F, L = 0.2 H, R = 800 \Omega$

Soln:

$$\frac{1}{LC} = 5 \times 10^6, \quad \frac{R^2}{4L^2} = 4 \times 10^6 < \frac{1}{LC}$$

$$\frac{1}{LC} > \frac{R^2}{4L^2}$$

$$\omega_0^2 > \frac{R^2}{4L^2}$$

Oscillatory.

$$\therefore \omega_1 = \sqrt{\omega_0^2 - \frac{R^2}{4L^2}}$$

$$2\pi f_1 = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$\therefore f_1 = \frac{1}{2\pi} \sqrt{5 \times 10^6 - 4 \times 10^6}$$

$$= \frac{10^3}{2\pi}$$

$$= 159.15 \text{ Hz}$$

Sitagil M

Exm $m = 0.5 \text{ kg}$, $K = 10.6 \text{ N/m}$,
 $b = 25$.

Soln: $\gamma = \frac{b}{m} = \frac{25}{0.5} = 50$

$\gamma^2 = 2500 \Rightarrow \frac{\gamma^2}{4} = 625$

$\omega_0 = \frac{K}{m} = \frac{10.6}{0.5} = 21.2$

$\therefore \omega_0^2 = 449.44 > \frac{\gamma^2}{4} = 625$

$\omega_0^2 > \frac{\gamma^2}{4}$; oscillatory.

Exm $m = 250 \text{ g}$, $K = 85 \text{ N/m}$, $b = 70 \text{ g/s}$

S.H.M, $\omega_0^2 = \frac{K}{m}$

D.H.M, $\omega_1 = \sqrt{\frac{K}{m} - \frac{b^2}{m^2}}$

